Using DroidLink AstroPixels

Complete setup, wiring, pairing, commands, servo calibration, and troubleshooting guide

> **What this guide covers**
>
> DroidLink AstroPixels is an ESP32-based dome controller for AstroPixels lighting, PCA9685 servo boards, holoprojectors, and optional Pololu Maestro controllers. It receives commands wirelessly from the DroidLink Master and can also pass compatible commands to devices connected to Serial2.

Important: AstroPixels uses reserved DroidLink Slave ID **7** and requires DroidLink Master firmware **V1.6 or newer**.

Guide revised for the current DroidLink AstroPixels firmware. Commands and hardware behavior were checked against the firmware source.

Contents

1. System overview

The DroidLink Master sends ESP-NOW command packets directly to AstroPixels. AstroPixels executes supported commands locally using the ReelTwo/Marcduino command engine. With Serial2 pass-through enabled by default, most received commands are also transmitted from the AstroPixels Serial2 TX pin.

```
DroidLink Display → DroidLink Master → ESP-NOW → AstroPixels → local lights/PCA servos
                                                          └→ Serial2 TX → downstream Marcduino-compatible device
```

What AstroPixels can control

- Front and rear logic displays
- Front and rear PSI lighting
- Front, rear, and top holoprojector lights and movement servos
- Dome panels through one or two PCA9685 servo boards
- Pololu Maestro scripted servo sequences over Serial2
- Additional Marcduino-compatible devices chained from Serial2 TX

> **Current build:** the normal AstroPixels Wi-Fi access point, web page, OTA, and Wi-Fi command receiver are disabled in this DroidLink build. Wireless control comes from DroidLink over ESP-NOW.

2. Before you begin

Required items

- DroidLink Master running firmware V1.6 or newer
- DroidLink AstroPixels ESP32 with current firmware
- A USB data cable and serial terminal for first-time setup
- Correctly powered AstroPixels LED hardware
- Optional PCA9685 board(s), Maestro controller(s), and servos

> **Servo safety:** test new servos without linkages attached. Set conservative limits before allowing full movement. Keep hands and tools clear whenever commands are being sent.

Information you will need

You need both wireless MAC addresses:

- **AstroPixels MAC:** printed in the USB serial console at boot. Add this to the Master as a slave.
- **DroidLink Master MAC:** paste this into the AstroPixels serial console during first-time setup.

3. Connections and wiring

Serial2 connector

Signal	ESP32 pin	Purpose
Serial2 RX	GPIO 16	Optional incoming serial command data
Serial2 TX	GPIO 17	Outgoing Marcduino text or Maestro binary commands
Ground	GND	Must be shared with the connected serial device

The default Serial2 rate is **9600 baud, 8 data bits, no parity, 1 stop bit**.

> **Do not connect two active transmitters to the same RX wire.** Confirm the signal direction before connecting AstroPixels, a Maestro, or another Marcduino-compatible controller. Use logic-level serial, not RS-232 voltage levels.

PCA9685 servo boards

AstroPixels is the I²C controller for the PCA9685 boards. Board 1 is used for dome panels. Board 2 is used for the six holoprojector movement axes. Power servos from an appropriate external supply and share ground with AstroPixels. Do not assume the ESP32 or USB connection can power the servos.

4. First-time setup and pairing

1. Connect AstroPixels to the computer by USB.
2. Open a serial terminal at **115200 baud**.
3. Power or reset AstroPixels. The console prints the AstroPixels MAC address and first-time instructions.
4. Open the DroidLink Master configuration and add the displayed AstroPixels MAC as a slave. AstroPixels always uses reserved Slave ID **7**.
5. Save the Master configuration. Do not reboot the Master yet.
6. Copy the DroidLink Master MAC address and paste it into the AstroPixels serial console in the form `AA:BB:CC:DD:EE:FF`.
7. AstroPixels saves the address and restarts.
8. When AstroPixels says it is waiting for the Master, reboot the Master within approximately 30 seconds.
9. After AstroPixels receives a packet from the saved Master, it stores the paired state and starts its lights, servos, and command engine.

> **Normal first-pair behavior:** the first packet establishes communication and starts AstroPixels. Send your test command again after the startup messages appear.

Status LED and console messages

- **Solid status LED during initial setup:** waiting for a Master MAC to be entered.
- **Triple pulse:** a command was received from the configured Master.
- **"No Master communication detected" after 30 seconds:** check both MAC addresses, save the Master configuration, and reboot the Master.

Start over with a different Master

At 115200 baud, type the following in the AstroPixels USB serial terminal and press Enter:

```
NEWMAC
```

This clears the stored Master MAC and paired state, then restarts AstroPixels. It does not describe a general factory reset of every AstroPixels preference.

5. How commands travel through AstroPixels

This is important when other devices are connected to Serial2.

Command type	Local action	Serial2 behavior
Normal AstroPixels/Marcduino command, such as `:OP00`	Executed locally by AstroPixels	Also sent as text followed by a carriage return when pass-through is enabled
Maestro command, such as `:D1S05`	Consumed by the Maestro adapter	Converted to a Pololu binary command; the original text is not forwarded
`:CAP?`	Requests AstroPixels capability information	Not treated as an ordinary device command

> **Example:** sending `:OP00` from DroidLink opens all locally controlled PCA panels and also sends `:OP00` plus a carriage return from Serial2 TX at 9600 baud. A downstream Marcduino-compatible device may therefore react to the same command.

> **Avoid accidental double actions.** If both AstroPixels and a downstream controller recognize the same command, both may move hardware. Disconnect the downstream serial device during initial PCA servo testing if you are unsure.

6. Testing basic commands

Use the command field in the DroidLink Display while it is in Configuration Mode. Send one command at a time and watch both the mechanism and USB serial output.

Command	Expected result
`@0T1`	Set both logic displays to normal
`@0T3`	Set both logic displays to alarm
`*ON01`	Turn on the front holoprojector light
`*OF01`	Turn off the front holoprojector light
`:CL00`	Close all PCA-controlled dome panels
`:OP00`	Open all PCA-controlled dome panels

> Only test `:OP00` after panel servo directions and safe endpoints are known. Opening every panel at once can damage a mechanism that is wired or calibrated incorrectly.

7. PCA9685 panel and servo control

Common panel commands

Command	Action
`:CL00`	Close all panels
`:OP00`	Open all panels
`:OF00`	Flutter all panels
`:OP01` through `:OP10`	Open panel group 1 through 10
`:CL01` through `:CL10`	Close panel group 1 through 10
`:OF01` through `:OF10`	Flutter panel group 1 through 10
`:OP11` / `:CL11`	Open / close the configured top pie-panel group
`:OP12` / `:CL12`	Open / close the configured lower dome-panel group

Direct servo tools

Format	Purpose
`:SQ<servo>,<pulse>`	Move one servo directly to a pulse value
`:SL<servo>,<minimum>,<maximum>`	Set and save servo endpoints
`:SM...`	Advanced direct movement using the ReelTwo servo-dispatch format
`:SF<easing>$<group>`	Set an easing method for a servo group

Persistent servo calibration

Every valid `:SL` command takes effect immediately and is saved automatically in ESP32 non-volatile storage. Saved values are restored at boot. No separate save command is required.

```
:SL0,900,2100
:SL18,1250,1750
```

Factory defaults are **800–2200**. Saved calibration normally survives a firmware upload, but it is lost if flash/NVS is erased.

8. Pololu Maestro control

AstroPixels can address three chained Maestro devices on Serial2:

DroidLink target	Maestro device ID	Example
D1	12	`:D1S05` runs script 5
D2	13	`:D2S05` runs script 5
D3	14	`:D3S05` runs script 5
Broadcast	12, 13, and 14	`:DS05` runs script 5 on all three

Script numbers must be between 0 and 99. Maestro commands are converted to Pololu binary protocol; they are not passed as ordinary Marcduino text.

> The Maestro must be configured for the same device ID used above and for the appropriate serial protocol. All devices must share ground.

9. Holoprojector installation and Auto Twitch

Default servo assignments

Holoprojector	PCA board	AstroPixels servo numbers
Front	Board 2	13 horizontal, 14 vertical
Top	Board 2	15 horizontal, 16 vertical
Rear	Board 2	17 vertical, 18 horizontal

> If the holoprojectors were previously connected to PCA Board 1, move them to Board 2 before using this default configuration.

Safe installation procedure

1. Remove or disconnect the mechanical linkages.
2. Center the front, rear, and top mechanisms:

```
*HPF101
*HPR102
*HPT103
```

3. Install each horn as straight as possible while the servo remains centered.
4. Attach the flexible linkages with the holoprojector physically centered.
5. Set conservative limits for servos 13 through 18:

```
:SL13,1200,1800
:SL14,1200,1800
:SL15,1200,1800
:SL16,1200,1800
:SL17,1200,1800
:SL18,1200,1800
```

Fine-tune one endpoint at a time. Reduce travel immediately if a servo buzzes, stalls, binds, or pushes into a stop.

Auto Twitch

Set the interval first; this does not start movement:

```
@HPA190,1,3
```

Target	Enable	Disable
All holoprojectors	`@HPA199`	`@HPA198`
Front	`@HPF199`	`@HPF198`
Rear	`@HPR199`	`@HPR198`
Top	`@HPT199`	`@HPT198`

Interval format:

```
@HP<target>190,<minimum seconds>,<maximum seconds>
```

Examples: `@HPF190,10,30`, `@HPR190,5,10`, and `@HPT190,15,25`.

10. Logic, PSI, holo-light, and sequence commands

Logic-display effects

The first digit selects the display: `1` front, `2` rear, and `0` both.

Example	Effect
`@0T1`	Normal
`@0T2`	Flashing color
`@0T3`	Alarm
`@0T4`	Failure
`@0T5`	Red-alert/scream effect
`@0T6`	Leia effect
`@0T11`	Imperial March effect

Text examples: `@1MHello` for the upper front logic, `@2MWorld` for the lower front logic, and `@3MAstromech` for the rear logic. Font commands include `@1P60` / `@2P60` / `@3P60` for Latin and `P61` variants for Aurebesh.

PSI effects

PSI commands follow a similar pattern: `@1P...` selects front PSI, `@2P...` selects rear PSI, and `@0P...` selects both. Common effect numbers are 1 normal, 2 flashing, 3 alarm, 4 failure, 5 red-alert, 6 Leia, and 11 March.

Holoprojector light and movement examples

Command	Action
`*ON01` / `*OF01`	Front holo light on / off
`*ON02` / `*OF02`	Rear holo light on / off
`*ON03` / `*OF03`	Top holo light on / off
`*ST00`	Reset all holoprojectors
`*RD01`, `*RD02`, `*RD03`	Random movement for front, rear, or top
`*HPS301` / `*HPS601`	Front holo pulse / rainbow; use final digit 2 or 3 for rear/top
`*HN01`, `*HN02`, `*HN03`	Nod front, rear, or top
`*HW01`, `*HW02`, `*HW03`	Wag front, rear, or top

AstroPixels extended logic effects

The `@APLE` command provides AstroPixels-specific effects. Its encoded value selects the target logic, effect, color, speed/sensitivity, and duration. Useful examples include:

Command	Result
`@APLE51000`	Solid red
`@APLE54000`	Solid green
`@APLE10500`	Default alarm
`@APLE20000`	Failure
`@APLE30008`	Leia effect for 8 seconds
`@APLE63315`	Flashing yellow for 15 seconds
`@APLE100500`	Rainbow
`@APLE225000`	Fire effect

11. Command-reference strategy

AstroPixels is intentionally compatible with a large portion of the Marcduino command language. The firmware contains far more commands than can be safely summarized in one setup guide, including complete sequences, advanced panel group animations, sound commands, and specialized logic effects.

When a command is not listed here:

1. Check the DroidLink Command Reference.
2. Check the AstroPixelsPlus documentation and command list.
3. Check a Marcduino command reference. PCA panel commands such as `:OP`, `:CL`, `:OF`, and many standard `:SE` sequences generally follow Marcduino conventions.
4. Test with linkages disconnected or with conservative endpoints. Compatibility does not guarantee that a command is mechanically safe for a particular build.

AstroPixelsPlus project: https://github.com/reeltwo/AstroPixelsPlus

> Commands sent through DroidLink should normally be entered exactly as documented, without adding a visible carriage return. DroidLink transports the command, and AstroPixels adds the required Serial2 terminator when passing it downstream.

12. Troubleshooting and recovery

Problem	Checks
AstroPixels remains in first-time setup	Enter the Master wireless MAC in the USB console. Use six hexadecimal pairs separated by colons.
No Master detected	Confirm the AstroPixels MAC is saved in the Master, confirm the correct Master MAC was entered into AstroPixels, save the Master configuration, then reboot the Master.
Paired previously but will not connect now	Verify that the same Master is being used. Type `NEWMAC` at 115200 baud to clear pairing and repeat setup.
First command after pairing did nothing	The first packet may have established pairing before the command engine started. Wait for startup to finish and send it again.
Local PCA panels move, but downstream device does not	Check Serial2 TX on GPIO 17, common ground, 9600 baud, downstream RX wiring, and downstream command compatibility.
Unexpected second device moves	The same normal command is executed locally and passed out Serial2. Disconnect or reconfigure the downstream device if both should not react.
Maestro does not run a script	Check device ID 12/13/14, shared ground, serial mode, script number 0–99, and use `:D1S`, `:D2S`, `:D3S`, or `:DS`.
Servo calibration disappeared	Check whether the installer performed a full flash/NVS erase. Normal uploads preserve calibration; erase operations do not.
Servo buzzes or binds	Stop movement, disconnect the linkage if necessary, and reduce the corresponding `:SL` range.
AstroPixels web page cannot be found	The current DroidLink build has the upstream AstroPixels Wi-Fi/web features disabled. Use DroidLink and the USB serial console.

13. Quick-reference page

Task	Command or setting
USB setup terminal	115200 baud
Serial2 output	GPIO 17, 9600 baud, 8N1
Reset Master pairing	`NEWMAC` in USB console
Open / close / flutter all panels	`:OP00` / `:CL00` / `:OF00`
Set servo limits	`:SL<servo>,<min>,<max>`
Run Maestro script 5 on D1	`:D1S05`
Run Maestro script 5 on all	`:DS05`
Normal / alarm / failure logics	`@0T1` / `@0T3` / `@0T4`
Front holo light on / off	`*ON01` / `*OF01`
Set all holo twitch interval	`@HPA190,5,10`
Enable / disable all holo twitch	`@HPA199` / `@HPA198`

> **Remember:** ordinary commands such as `:OP00` execute locally and are also passed out Serial2 by default. Maestro `:D...` commands are converted to binary Maestro instructions instead.

End of guide.